

M KARTHIKEYAN

AI Engineer | Machine Learning Specialist | OT Security Analytics

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🌐 **Portfolio:** karthikeyan1708.github.io — *Interactive portfolio with projects, engagements & experience*

Professional Summary

Results-driven AI Engineer with **1.8+ years** at a Big-4 consulting firm, specializing in building production-grade machine learning pipelines for industrial cybersecurity. Expertise in designing anomaly-detection models, automating large-scale PCAP analysis, and deploying AI-powered vulnerability assessment tools across critical infrastructure. Published researcher in ML-driven disease prediction. Proven track record of reducing manual security analysis effort by 40% through intelligent automation.

Core Competencies

- **Machine Learning & AI:** Supervised/Unsupervised Learning, Time-Series Forecasting, Anomaly Detection, Transfer Learning, Ensemble Methods (XGBoost, Random Forest), Hyperparameter Tuning
- **Deep Learning:** CNN, RNN, LSTM, Transformer Architectures, Whisper (ASR), Attention Mechanisms
- **Frameworks & Libraries:** TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face, ONNX
- **Data Engineering:** Python (Advanced), NumPy, Pandas, Dask, SQL, MongoDB, Feature Engineering, ETL Pipelines
- **Visualization & Reporting:** Matplotlib, Seaborn, Plotly, Power BI, Jupyter Notebooks
- **MLOps & DevOps:** Docker, Git, CI/CD (GitHub Actions), FastAPI, REST APIs, Linux
- **Cloud & Infrastructure:** AWS (S3, SageMaker, Lambda), Azure (Sentinel, Blob Storage)
- **Security Domain:** Wireshark, Claroty, Tenable, PCAP Analysis, OT/ICS Protocol Analysis

Professional Experience

Associate – AI & OT Cybersecurity

Oct 2024 – Present

PricewaterhouseCoopers (PwC), Hyderabad

1.8+ years

AI-Driven Security Analytics & Automation

- Architected and deployed an **ML-based anomaly detection pipeline** processing 500K+ OT sensor events daily, achieving 85% precision in identifying malicious traffic patterns and reducing false-positive rates by 30%.
- Engineered automated Python workflows to ingest, parse, and analyze **multi-terabyte PCAP datasets**, extracting 50+ protocol-level features for downstream classification models.
- Developed an **XGBoost-based risk-scoring engine** that prioritized vulnerabilities by exploitability and asset criticality, cutting manual triage time by **40%** across assessment engagements.
- Built reusable **NLP pipelines** for automated policy-document analysis, extracting compliance gaps against ISO 27001:2022 controls with 90%+ extraction accuracy.

Assessment Delivery & Technical Leadership

- Spearheaded OT security assessments for **6 major engagements** spanning aviation, energy, telecom, and manufacturing sectors – each involving network architecture review, stakeholder workshops, and executive-level reporting.
- Led on-site audits across **4 geographically distributed sites** under aggressive 3-week timelines, delivering comprehensive maturity assessments and remediation roadmaps.
- Performed deep-dive **firewall rule analysis and network segmentation reviews** for industrial control environments, identifying 25+ critical misconfigurations per engagement.
- Conducted **physical and logical security reviews** of telecom infrastructure across 5 tower locations, automating configuration compliance checks with custom Python scripts.
- Managed full project lifecycle for a multi-phase security engagement – from kickoff through final closure, including stakeholder coordination, milestone tracking, and payment reconciliation.

SOC Operations & Continuous Monitoring

- Currently supporting a **24/7 shift-based SOC operation** for a global industrial technology provider, performing real-time threat monitoring, alert triage, and incident escalation using Microsoft Sentinel and Defender for IoT.

Key AI/ML Projects

Automatic Speech Recognition & Emotion Detection – Tamil Language Apr–May 2024

- Fine-tuned OpenAI’s **Whisper Base** model using transfer learning on a custom Tamil-speech corpus; appended bidirectional LSTM layers for temporal emotion modeling, achieving **85% classification accuracy** across 6 emotion categories.
- Designed a robust data pipeline handling audio augmentation (noise injection, pitch shifting, time stretching), MFCC/spectrogram extraction, and batch processing of 12+ hours of raw audio daily.
- Deployed inference endpoint via FastAPI with sub-200ms latency; containerized with Docker for reproducible deployment.

Communicable Disease Outbreak Prediction System Jan–Mar 2023

- Developed an ensemble of **RNN and SVR models** to forecast dengue and chikungunya outbreaks with **92% accuracy** over a 12-month prediction horizon.
- Engineered 30+ features from epidemiological, meteorological, and demographic datasets totaling 2M+ records; applied PCA and recursive feature elimination for dimensionality reduction.
- Built interactive Plotly dashboards for public-health stakeholders, enabling drill-down analysis by region, time period, and disease type.
- **Published** in Springer – *Inventive Communication and Computational Technologies* (Oct 2023).

Additional Contributions

- **Proposal Development:** Co-authored technical proposals for two enterprise clients in FMCG and industrial sectors, contributing solution architecture and AI-enablement strategy sections.
- **Knowledge Sharing:** Created internal playbooks on ML-enhanced PCAP analysis and automated compliance checking, adopted by 3 project teams.

Certifications

ISO 27001:2022 Lead Implementer – Information Security Management Systems (ISMS)

Publications

M Karthikeyan et al., “*Communicable Disease Prediction Using Machine Learning and Deep Learning Algorithms*,” *Inventive Communication and Computational Technologies*, pp. 979–992, Springer, Oct 2023.
DOI: [10.1007/978-981-99-5166-6_66](https://doi.org/10.1007/978-981-99-5166-6_66)

Education

B.Tech, Computer Science & Engineering (Artificial Intelligence) 2020–2024
School of AI & Computing, Amrita Vishwa Vidyapeetham, Coimbatore CGPA: 8.5 / 10
Key Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Data Structures & Algorithms, Software Engineering, Database Management

Languages

English (Professional) | Tamil (Native) | Hindi (Conversational) | Kannada (Conversational)